

Circuit Analysis (Lab11)

Experiment 10: Verification of Norton's Theorems

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Date: Grade and Signature:

CLO2 CLO3	Use with identification the circuit components, breadboards, multimeter, power supplies, signal generators, and oscilloscopes. Make measurements in their own constructed electric circuits.			
Psychomotor/Affective	Level1 (1)	Level2 (2-3)	Level3 (4-5)	Level4 (6-7)
Report Marks (3)				
	Total marks (10)			

Objective

The objective of this lab is verify Norton's theorem and to find the full load current.

Apparatus Required

Sr. no	Apparatus	Range	Quantity
1	Regulated Power Supply	Standard	1
2	Signal Generator	Standard	1
3	Oscilloscope	Standard	1
4	Digital Multimeter	Standard	1
5	Breadboard & Wires	Standard	As Required
6	Resistors	1k, 330 ohms	1,3
7	Potentiometer	Standard	1

Description

Statement:

Any linear circuit can be simplified to an equivalent circuit consisting of a single current source and parallel resistance that is connected to a load.

Precautions:

1. Voltage control knob of RPS should be kept at minimum position.
2. Current control knob of RPS should be kept at maximum position

Procedure for Norton's Theorem:

1. Connections are given as per circuit diagram.
2. Set a particular value in RPS and note down the ammeter readings in the original circuit.

To Find I_N :

3. Remove the load resistance and short circuit the terminals.
4. For the same RPS voltage note down the ammeter readings.

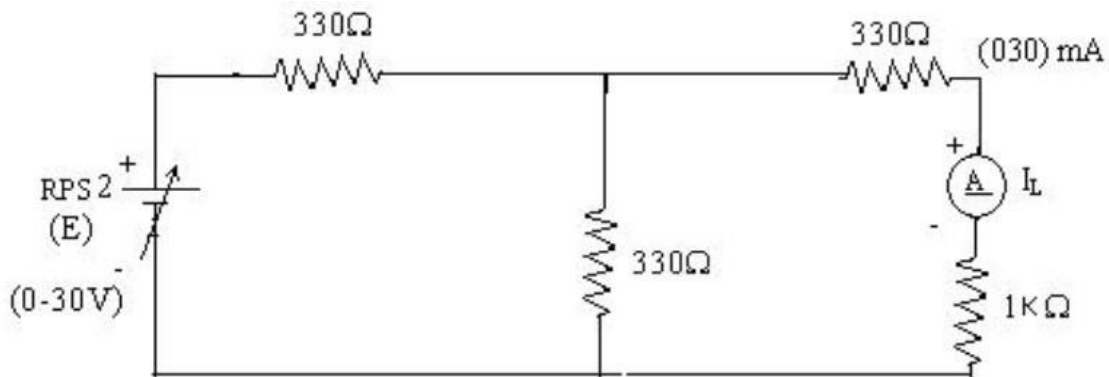
To Find R_N :

5. Remove RPS and short circuit the terminal and remove the load and note down the resistance across the two terminals.

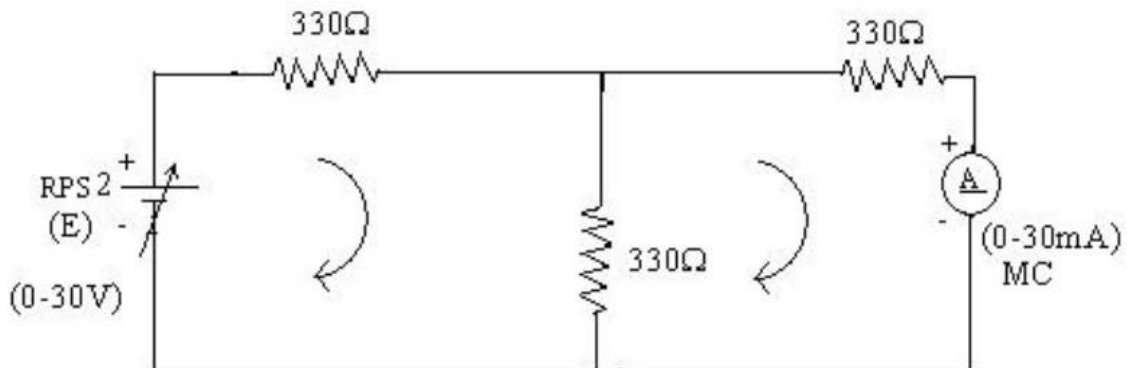
Equivalent Circuit:

6. Set I_N and R_N and note down the ammeter readings.
7. Verify Norton's theorem.

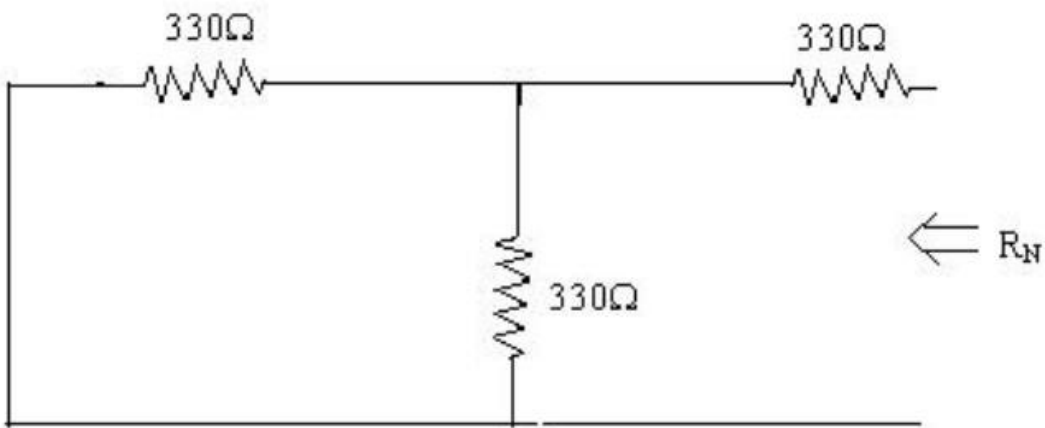
To find load current in circuit 1:



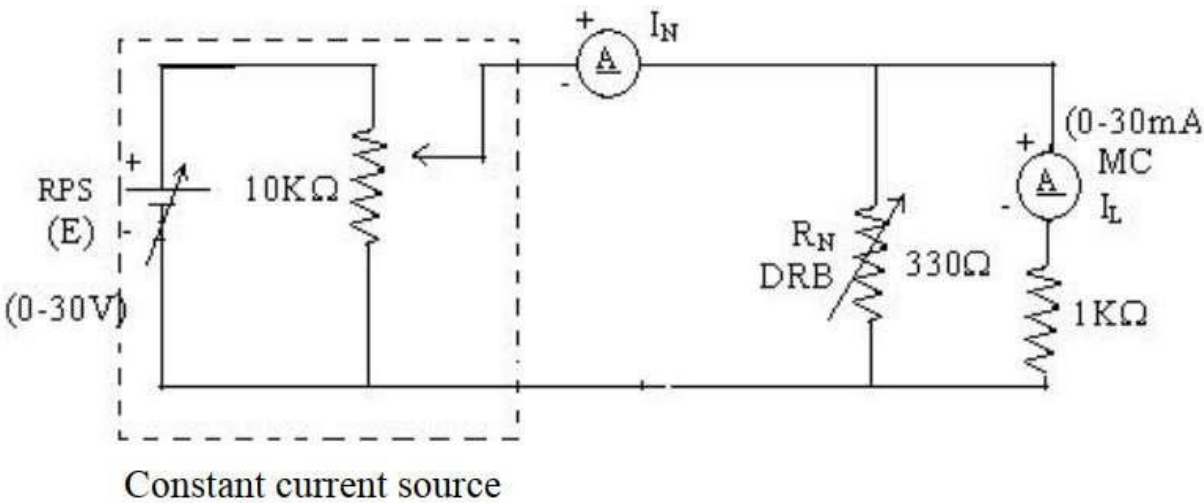
To find I_N



To find R_N



Norton's equivalent circuit



Norton's Practical Values

	E (volts)	I_N (mA)	R_N (ohms)	I_L (mA)	
				Circuit - I	Equivalent Circuit
Practical Values	10V	1mA	330 ohm	0.25mA	0.25mA

Post Lab Questions

1. What is Norton's Theorem?

Norton's Theorem states that any linear electrical circuit containing voltage sources, current sources, and resistors can be replaced by an equivalent circuit consisting of:

1. a single current source (I_n) 2. connected in parallel with a single resistor (R_n)
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1. Remove the load resistor from the circuit.

2. Find the Norton current and resistance.

3. Draw the equivalent circuit with: current source in parallel with resistor

4. Reconnect the load resistor.

2. What are the applications of Norton's Theorem?

1. It is used to simplify complex electrical circuits.

2. It helps in analyzing load current and load voltage easily.

3. It is used in circuit design and network analysis.

4. It is useful for studying parallel circuits.

5. It saves time in solving large electrical networks.

6. It is commonly used in electronic and power systems.

3. What are the theoretical and practical limitations of Norton's Theorem?

1. Norton's Theorem is applicable only to linear and bilateral circuits.

2. It cannot be used directly for non-linear elements such as diodes and transistors.

3. It is not suitable for circuits containing only dependent sources without independent sources.

4. The theorem becomes difficult for very complex circuits..

5. Practical measurements may contain errors due to internal resistance and instruments limitations.

6. High-frequency circuits may not give accurate results because of parasitic effects
